

Computer Networks (4th Edition) -Andrew S. Tanenbaum



Chapter 1

Introduction



1.1 Uses of Computer Networks

- Business Applications
- Home Applications
- Mobile Users
- Social Issues



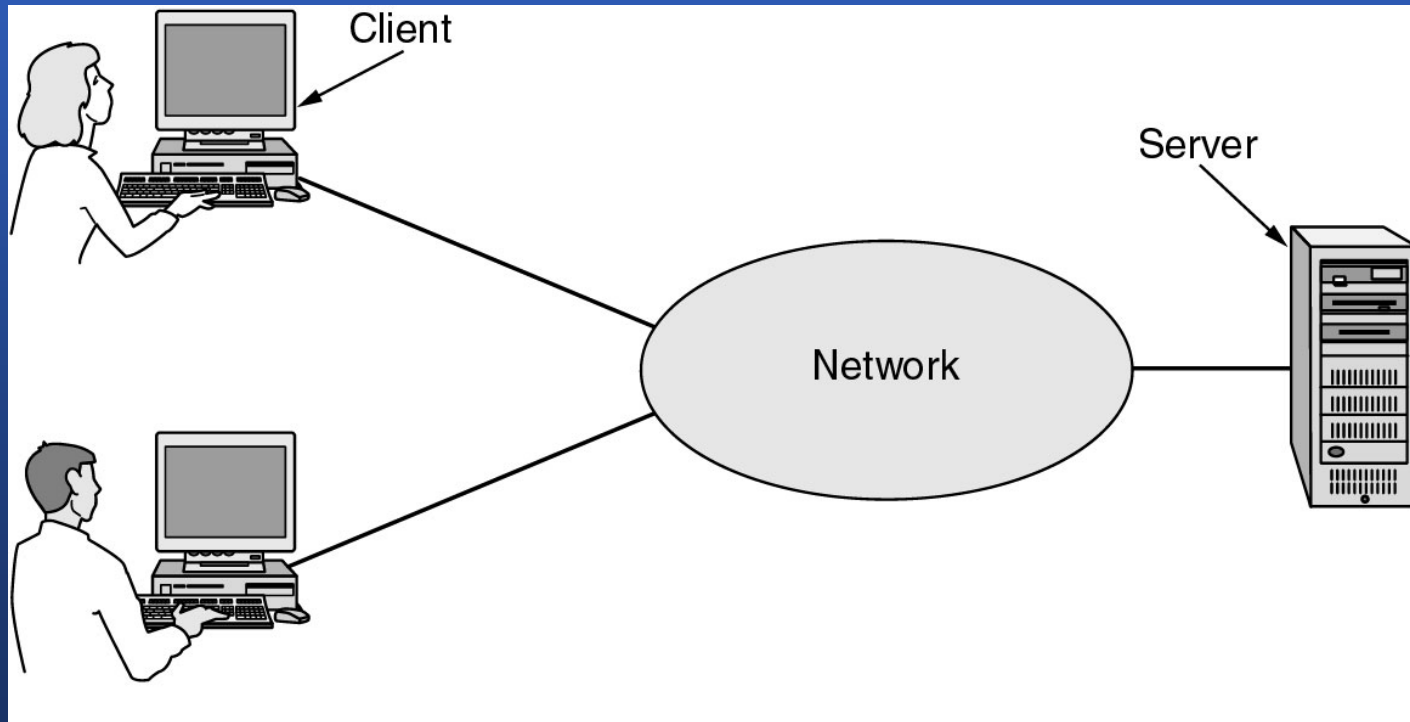
1.1.1 Business Applications of Networks

- 资源共享 (Resource Sharing)
- 高可靠性 (High Reliability)
- 节约经费 (Saving Investment)



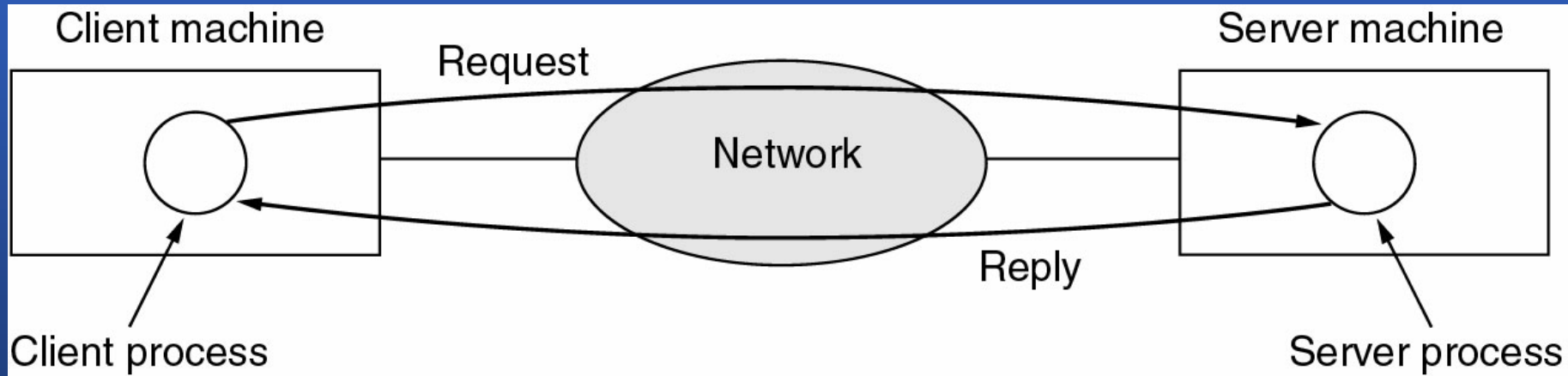
Business Applications of Networks (2)

- A network with two clients and one server.



Business Applications of Networks (3)

- The client-server model involves requests and replies.



Business Applications of Networks (4)

- Email
- Ftp
- Web
- Videoconference
- E-commence



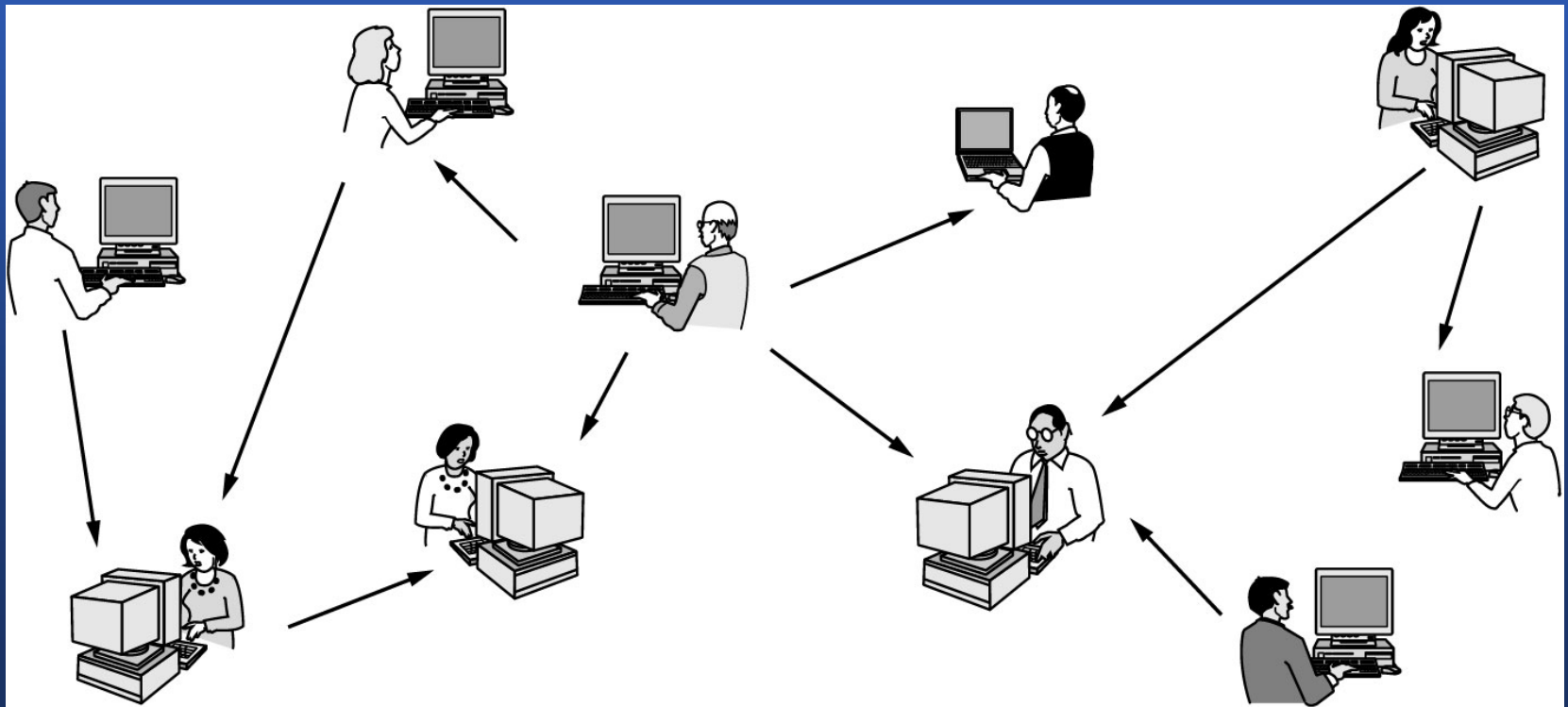
1.1.2 Home Network Applications

- Access to remote information
- Person-to-person communication
- Interactive entertainment
- Electronic commerce



Home Network Applications (2)

- In peer-to-peer system there are no fixed clients and servers.



Home Network Applications (3)

- Some forms of e-commerce.

Tag	Full name	Example
B2C	Business-to-consumer	Ordering books on-line
B2B	Business-to-business	Car manufacturer ordering tires from supplier
G2C	Government-to-consumer	Government distributing tax forms electronically
C2C	Consumer-to-consumer	Auctioning second-hand products on-line
P2P	Peer-to-peer	File sharing



1.1.3 Mobile Network Users

- Combinations of wireless networks and mobile computing.

Wireless	Mobile	Applications
No	No	Desktop computers in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in older, unwired buildings
Yes	Yes	Portable office; PDA for store inventory



1.1.4 Social Issues

- Politics
- Religion
- Security
- Sex
- Employer vs. employee
- Government vs. citizen



1.2 Network Hardware

- Local Area Networks
- Metropolitan Area Networks
- Wide Area Networks
- Wireless Networks
- Home Networks
- Internetworks



Broadcast Networks

- Types of transmission technology
- Broadcast links
- Point-to-point links
 - Unicasting

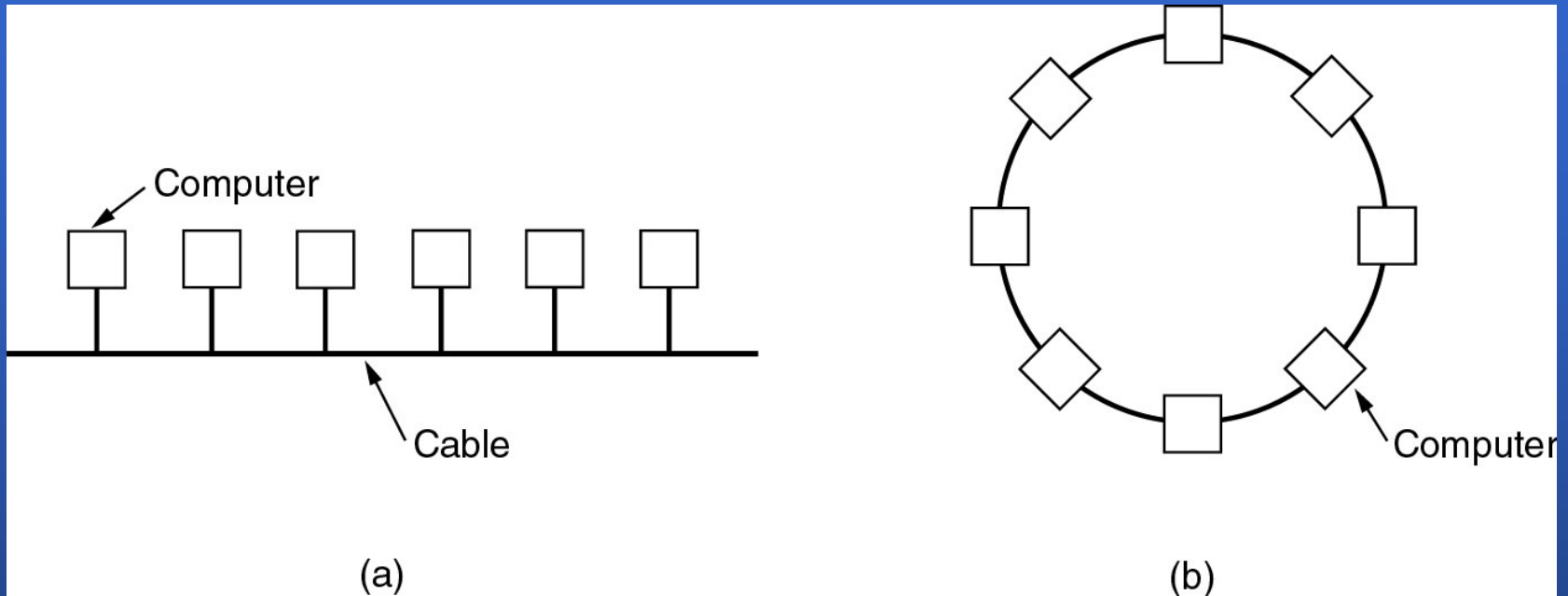


Broadcast Networks (2)

- Classification of interconnected processors by scale.

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	
100 m	Building	
1 km	Campus	
10 km	City	Local area network
100 km	Country	
1000 km	Continent	Metropolitan area network
10,000 km	Planet	
		Wide area network
		The Internet

1.2.1 Local Area Networks



- IEEE 802.3 Ethernet
- 10Mbps – 10Gbps
- IEEE 802.5 (IBM, 4/16Mbps)

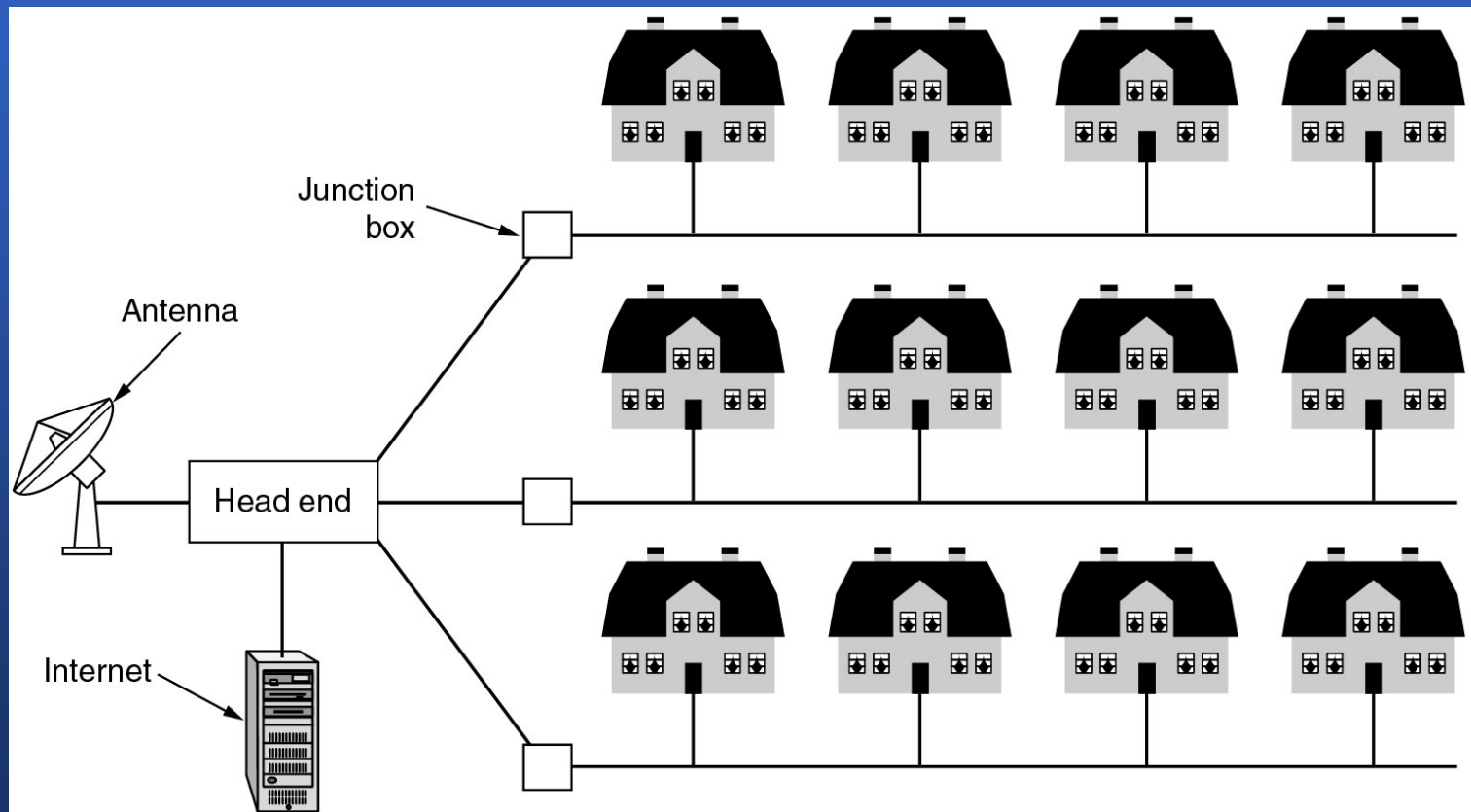
- Two broadcast networks

(a) Bus

(b) Ring

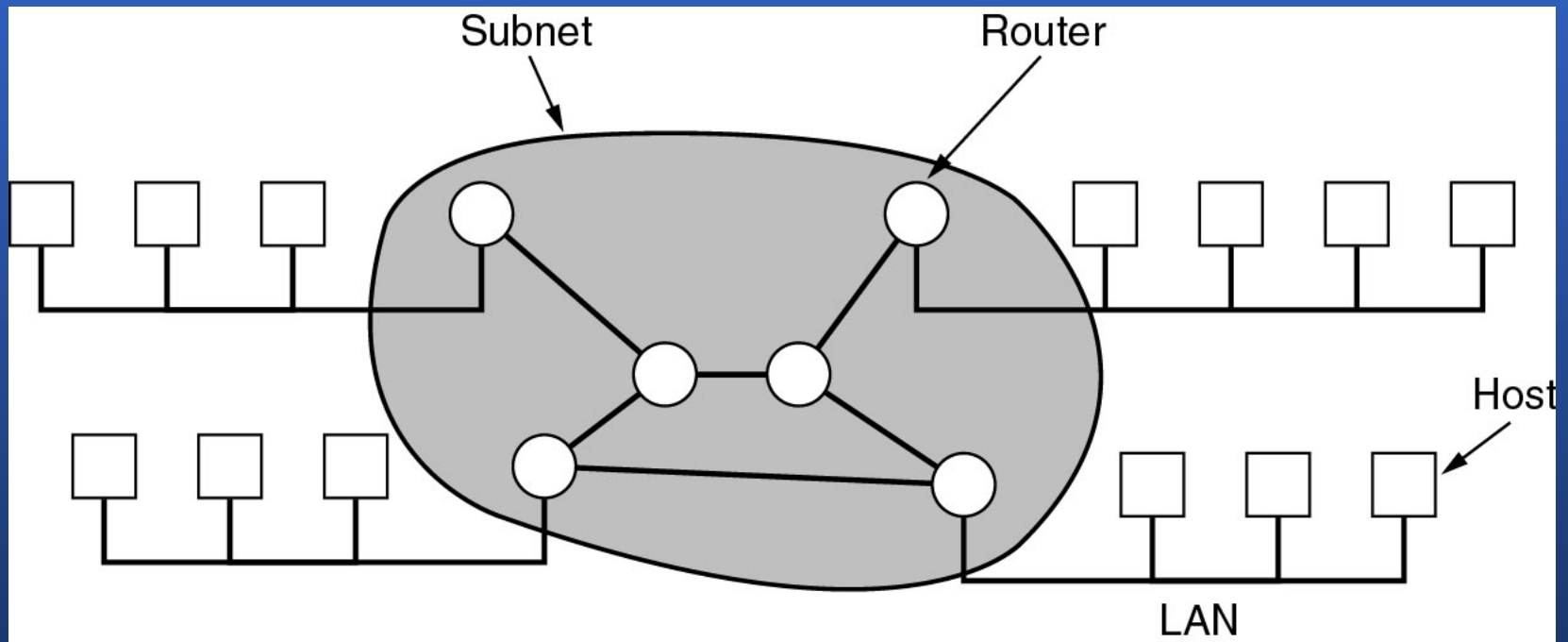
1.2.2 Metropolitan Area Networks

- A metropolitan area network based on cable TV.



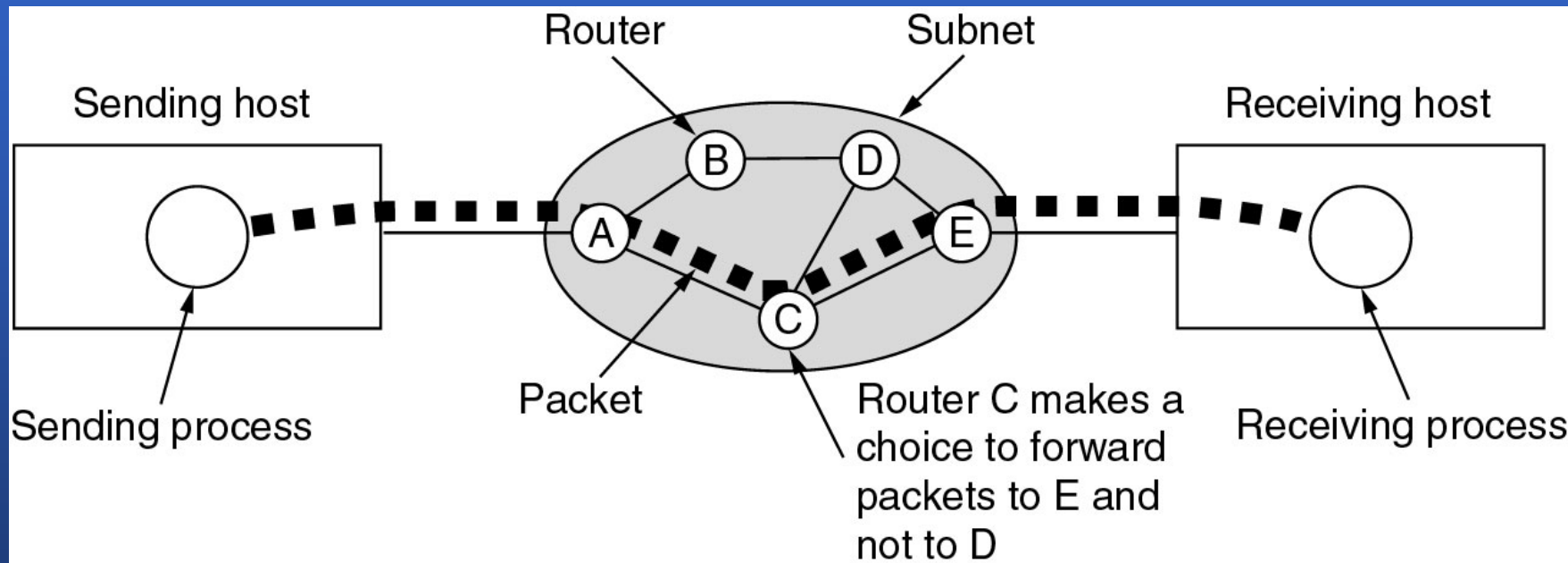
1.2.3 Wide Area Networks

- Relation between hosts on LANs and the subnet.



Wide Area Networks (2)

- A stream of packets from sender to receiver.



•Router

•Store-and-forward

•Packet switching



1.2.4 Wireless Networks

- Categories of wireless networks:
- System interconnection
- Wireless LANs
 - IEEE 802.11(.11a,.11b,.11g,.11i...)
 - WAPI (无线局域网鉴别与保密基本结构)
- Wireless WANs
 - IEEE 802.16



Wireless Networks (2)



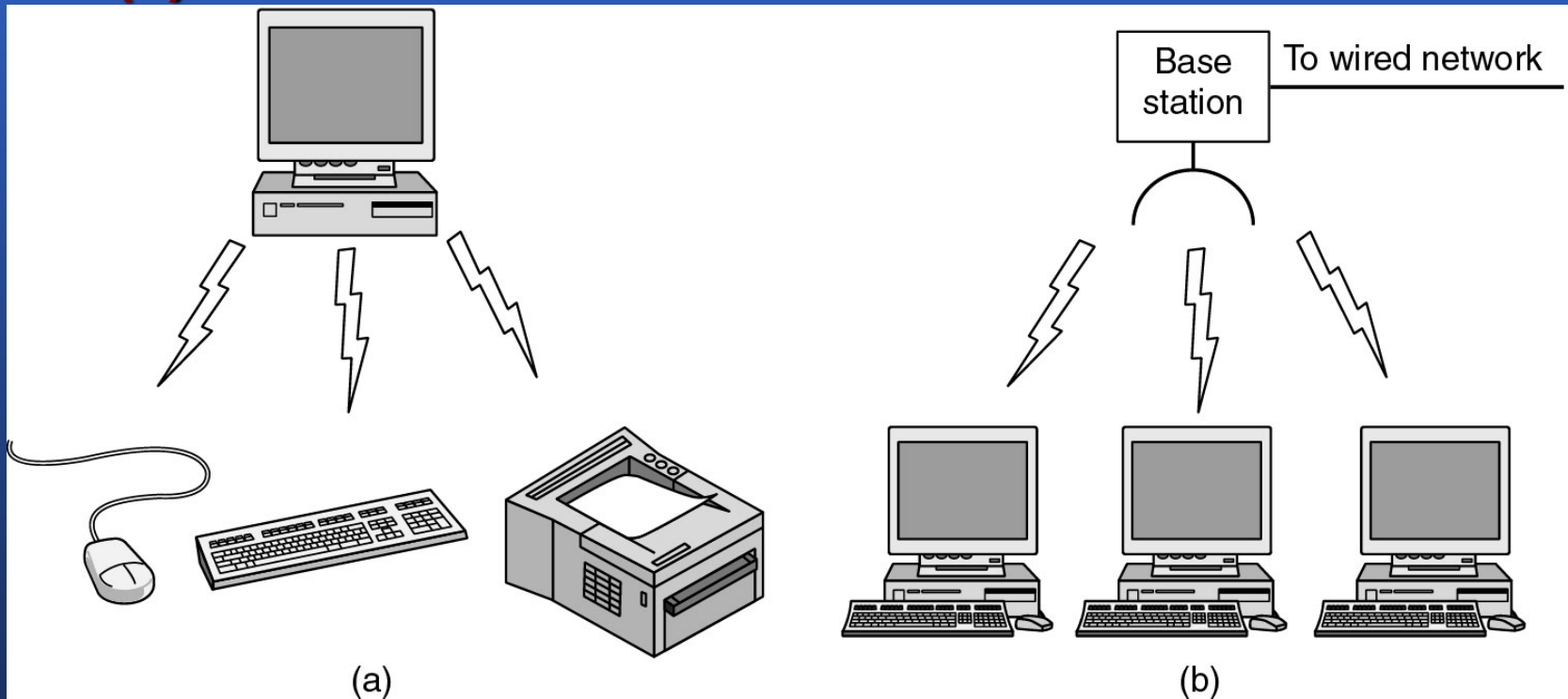
美国多层次联合封杀 WAPI 标准



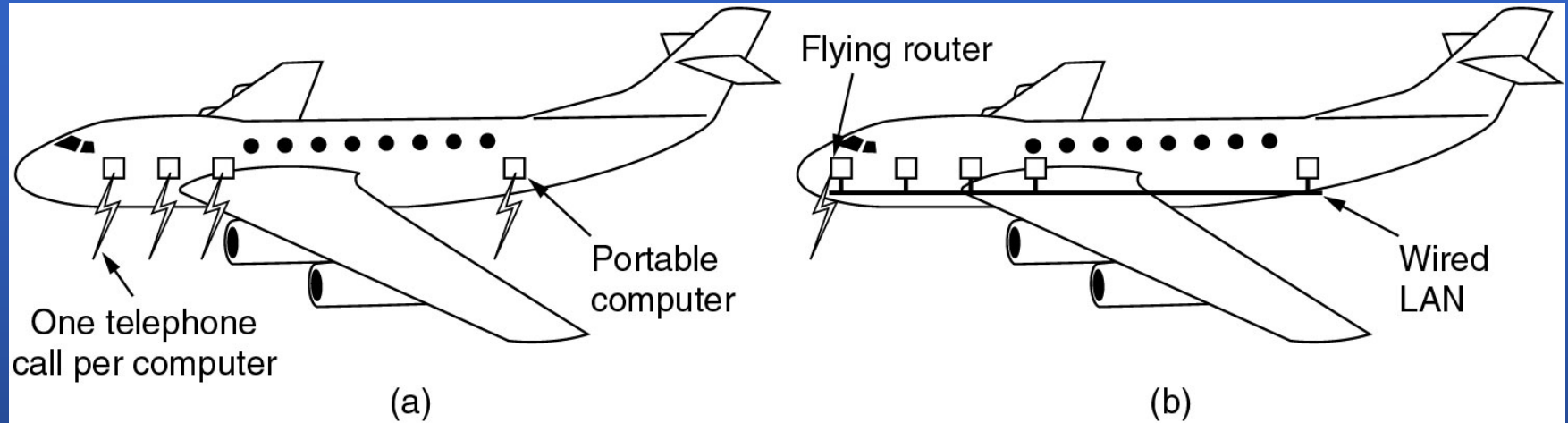
Wireless Networks (3)

(a) Bluetooth configuration

(b) Wireless LAN



Wireless Networks (4)



(a) Individual mobile computers

(b) A flying LAN

1.2.5 Home Network Categories

- Computers (desktop PC, PDA, shared peripherals)
- Entertainment (TV, DVD, VCR, camera, stereo, MP3)
- Telecomm (telephone, cell phone, intercom, fax)
- Appliances (microwave, fridge, clock, furnace, airco)
- Telemetry (utility meter, burglar alarm, babycam).



1.2.6 Internetworks

- Gateways
- Internetwork or internet

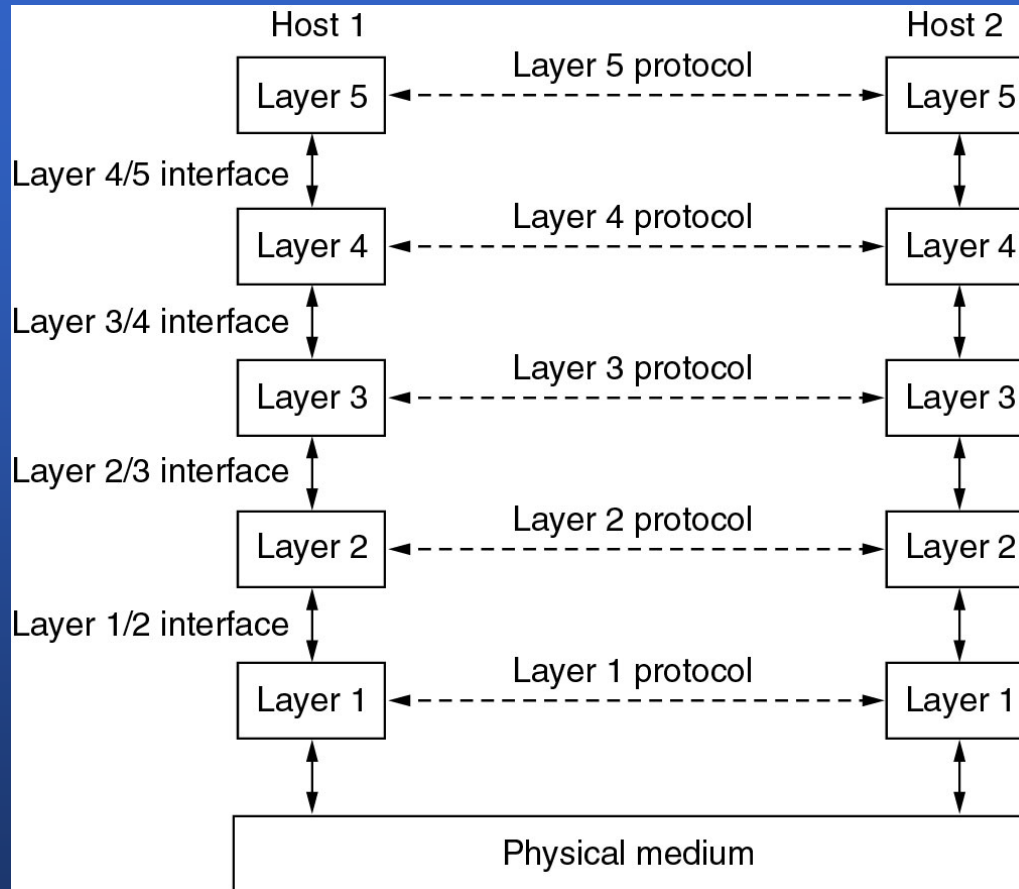


1.3 Network Software

- Protocol Hierarchies
- Design Issues for the Layers
- Connection-Oriented and Connectionless Services
- Service Primitives
- The Relationship of Services to Protocols



1.3.1 Network Software Protocol Hierarchies

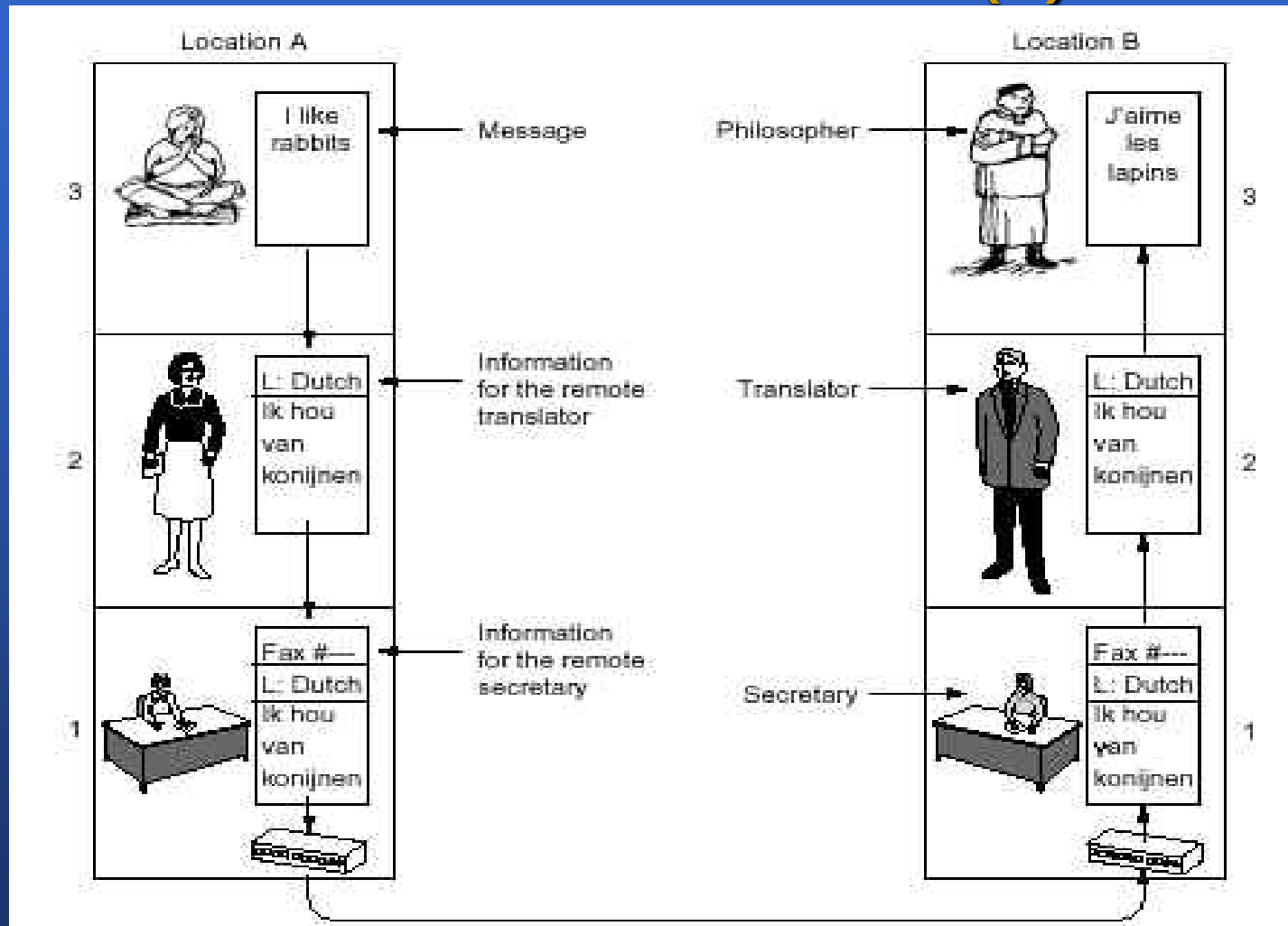


Layers, protocols, and interfaces.

Protocol Hierarchies (2)

- Urdu
- English

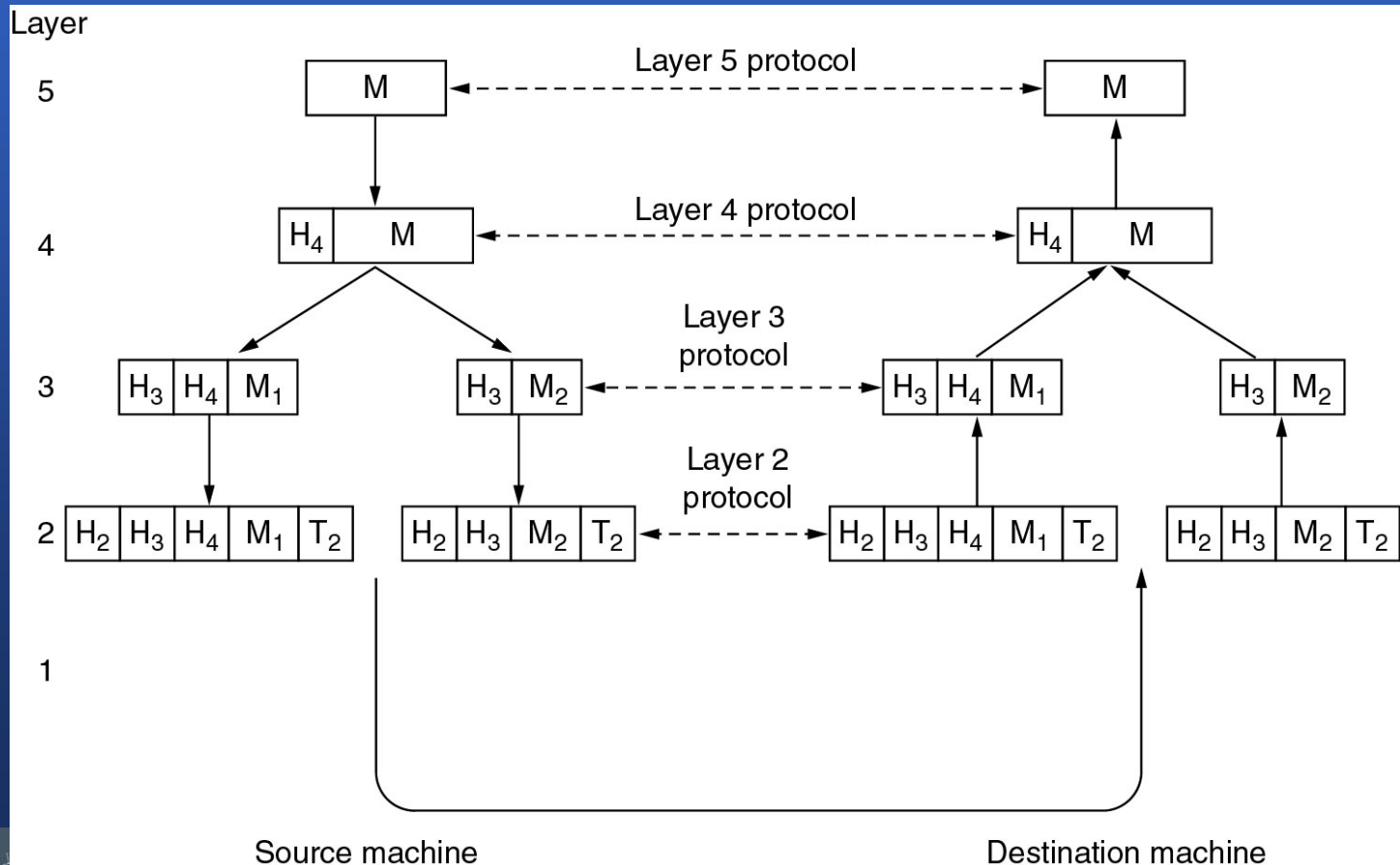
- Chinese
- French



The philosopher-translator-secretary architecture.

Protocol Hierarchies (3)

- Example information flow supporting virtual communication in layer 5.



1.3.2 Design Issues for the Layers

- Addressing
- Error Control
- Flow Control
- Multiplexing
- Routing



1.3.3 Connection-Oriented and Connectionless Services

- Six different types of service.

		Service	Example
		Reliable message stream	Sequence of pages
Connection-oriented	{	Reliable byte stream	Remote login
		Unreliable connection	Digitized voice
		Unreliable datagram	Electronic junk mail
Connection-less	{	Acknowledged datagram	Registered mail
		Request-reply	Database query

- Negotiation
- Quality of service

1.3.4 Service Primitives

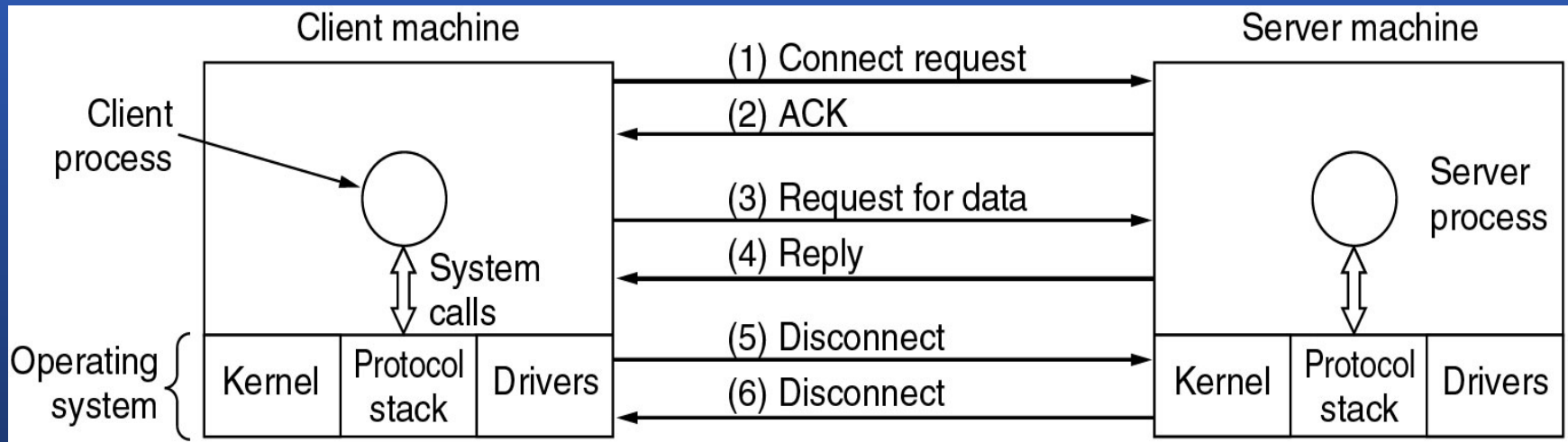
- Five service primitives for implementing a simple connection-oriented service.

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection



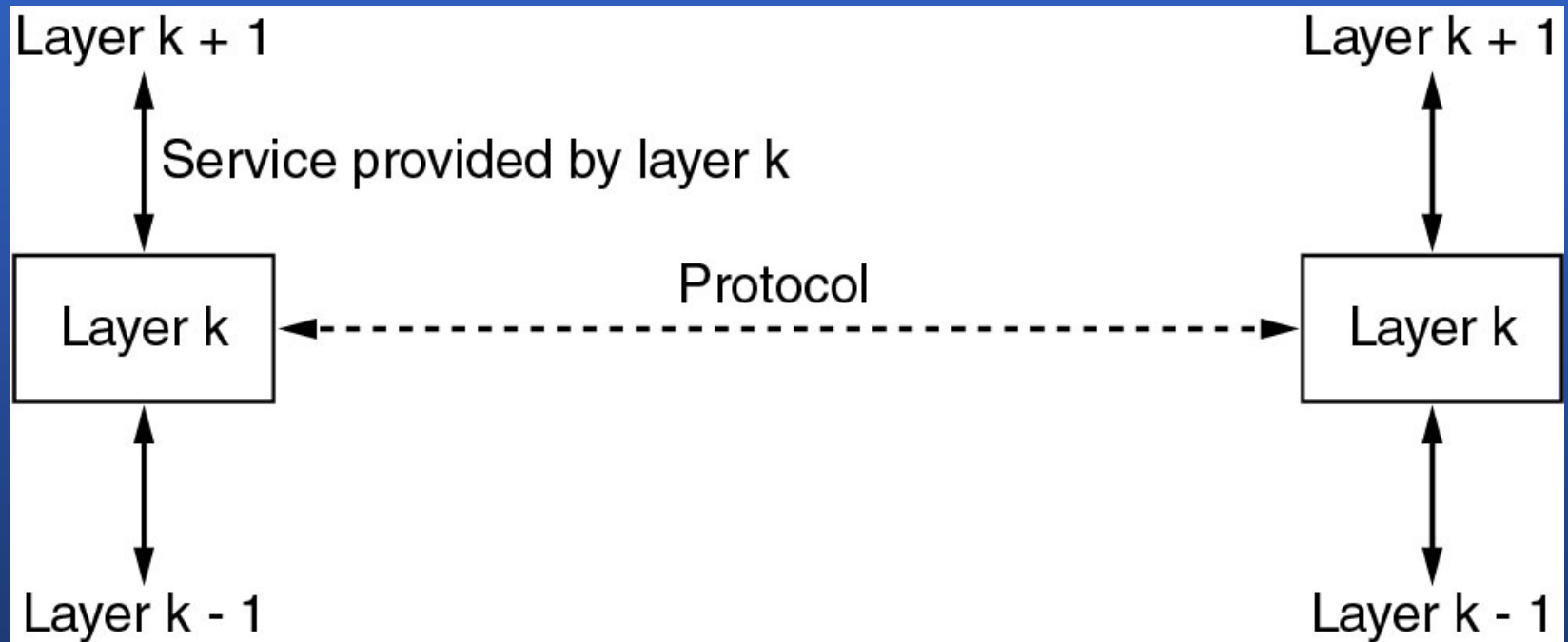
Service Primitives (2)

- Packets sent in a simple client-server interaction on a connection-oriented network.



1.3.5 Services to Protocols Relationship

- The relationship between a service and a protocol.



1.3.5 Services to Protocols Relationship

- Services:
 - 各层向它上层提供的一组原语（操作）
- Protocols:
 - 定义同层对等实体之间交换的帧、分组和报文的格式及意义的一组规则



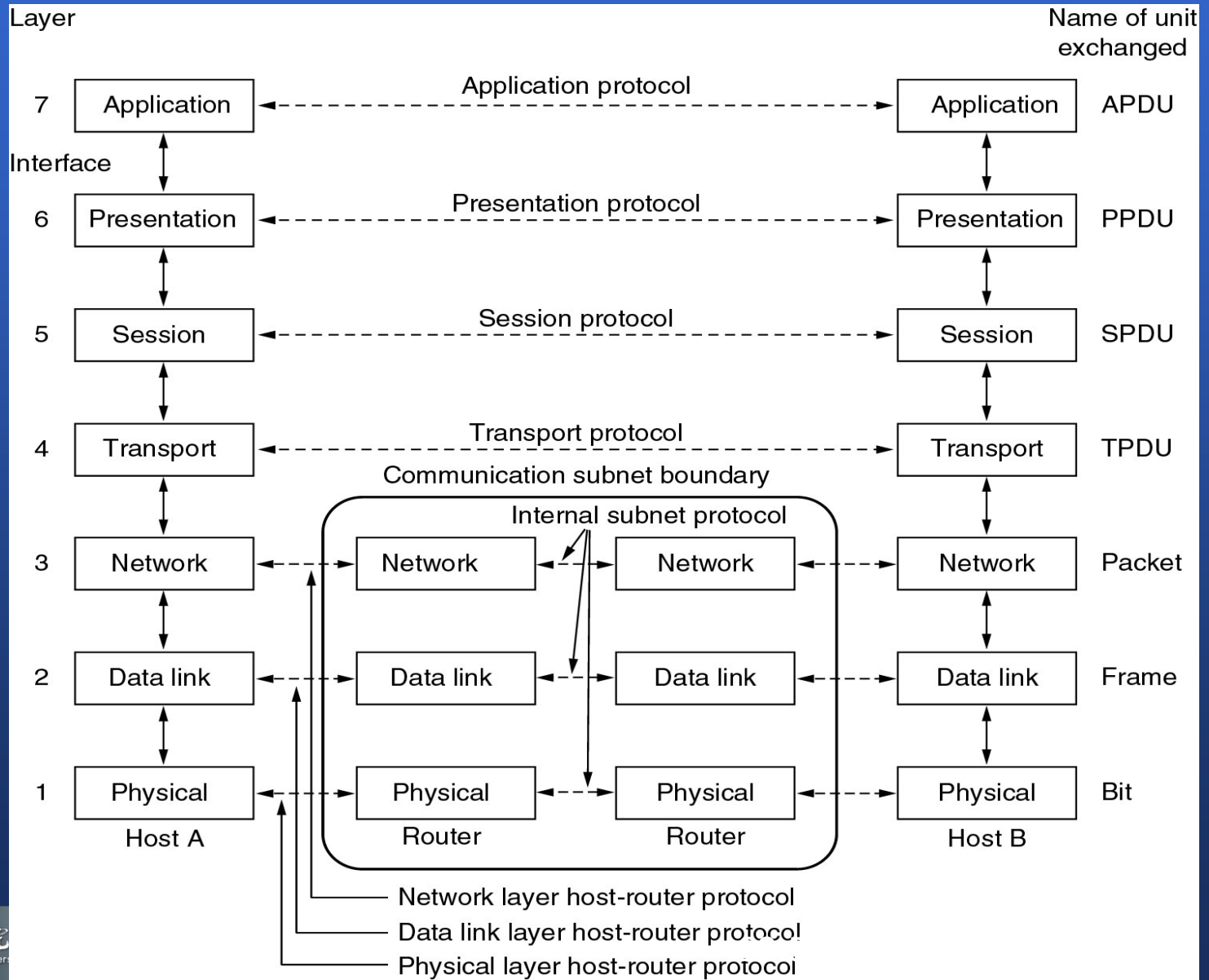
1.4 Reference Models

- The OSI Reference Model
- The TCP/IP Reference Model
- A Comparison of OSI and TCP/IP
- A Critique of the OSI Model and Protocols
- A Critique of the TCP/IP Reference Model



1.4.1 The OSI Reference Models

The OSI reference model.



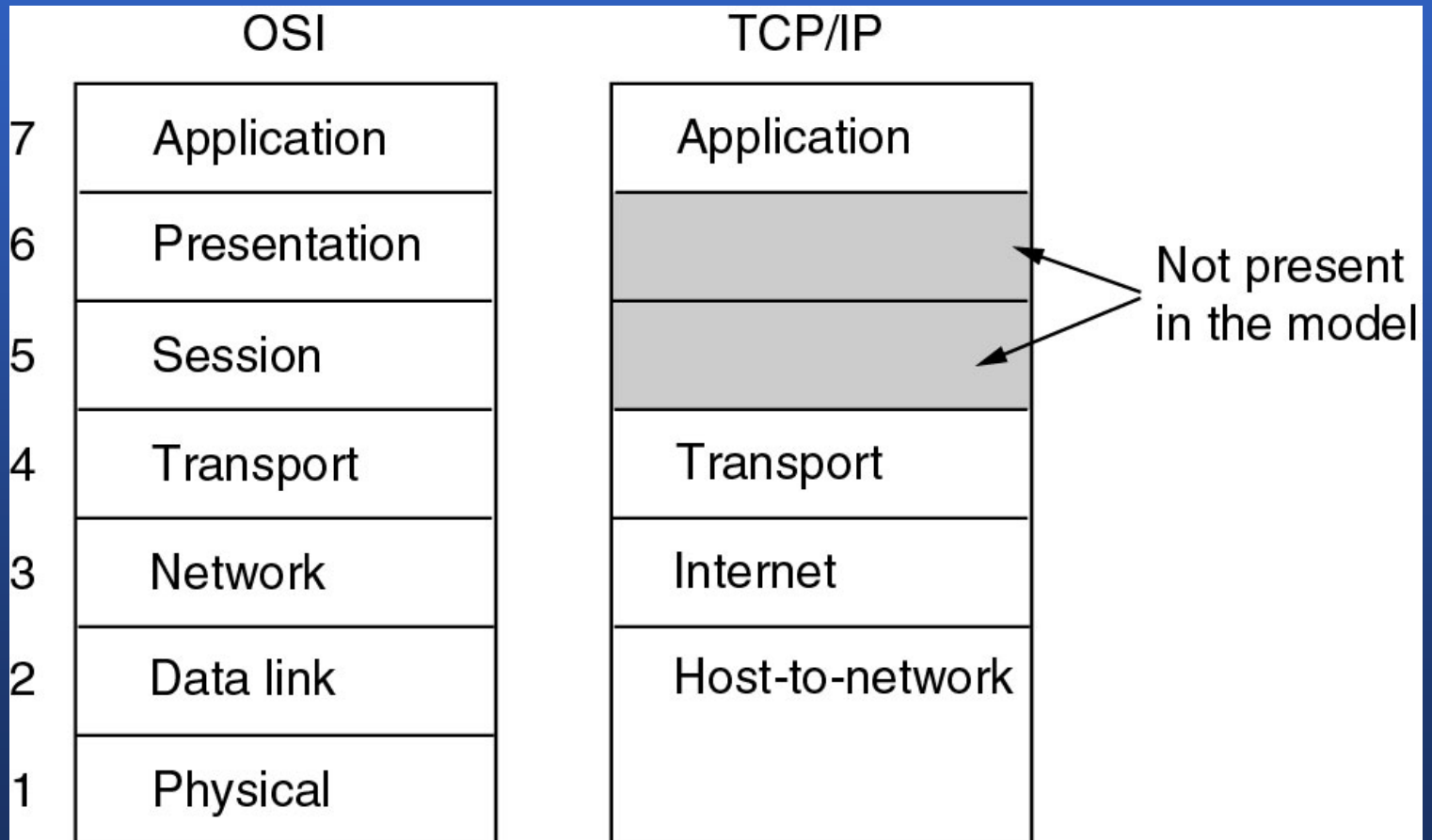
The OSI Reference Models

- Physical layer
- Data link layer
- Network layer
- Transport layer
- Session layer
- Presentation layer
- Application layer



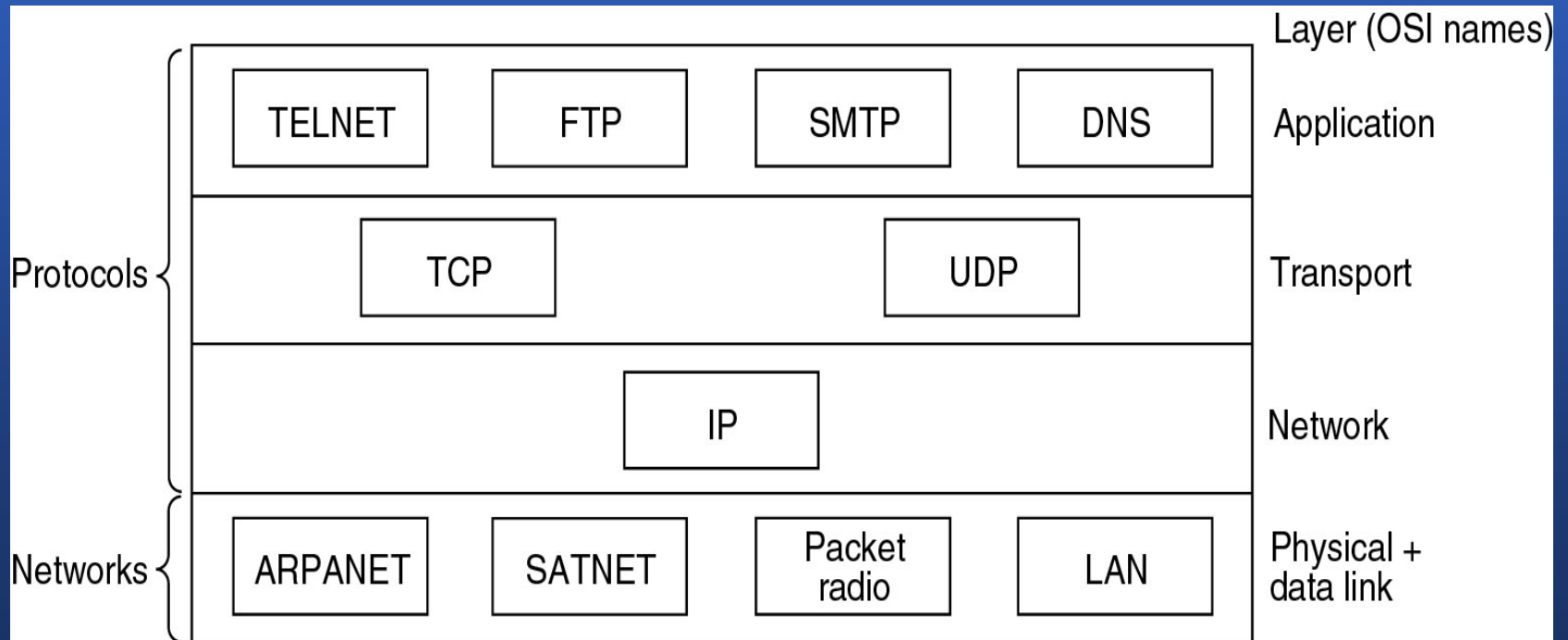
1.4.2 The TCP/IP Reference Models

- The TCP/IP reference model.



Reference Models (3)

- Protocols and networks in the TCP/IP model initially.



1.4.3 Comparing OSI and TCP/IP Models

- Concepts central to the OSI model
- Services
- Interfaces
- Protocols



OSI参考模型vs TCP/IP参考模型

●OSI :

- 3个主要概念：服务、接口、协议
- 协议有很好的隐藏性
- 产生在协议发明之前
- 共有7层
 - 网络层：连接和无连接
 - 传输层：面向连接

●TCP/IP

- 没有明确区分：服务、接口、协议
- 产生在协议发明之后
- 共有5层
 - 网络层：无连接
 - 传输层：面向连接和无连接



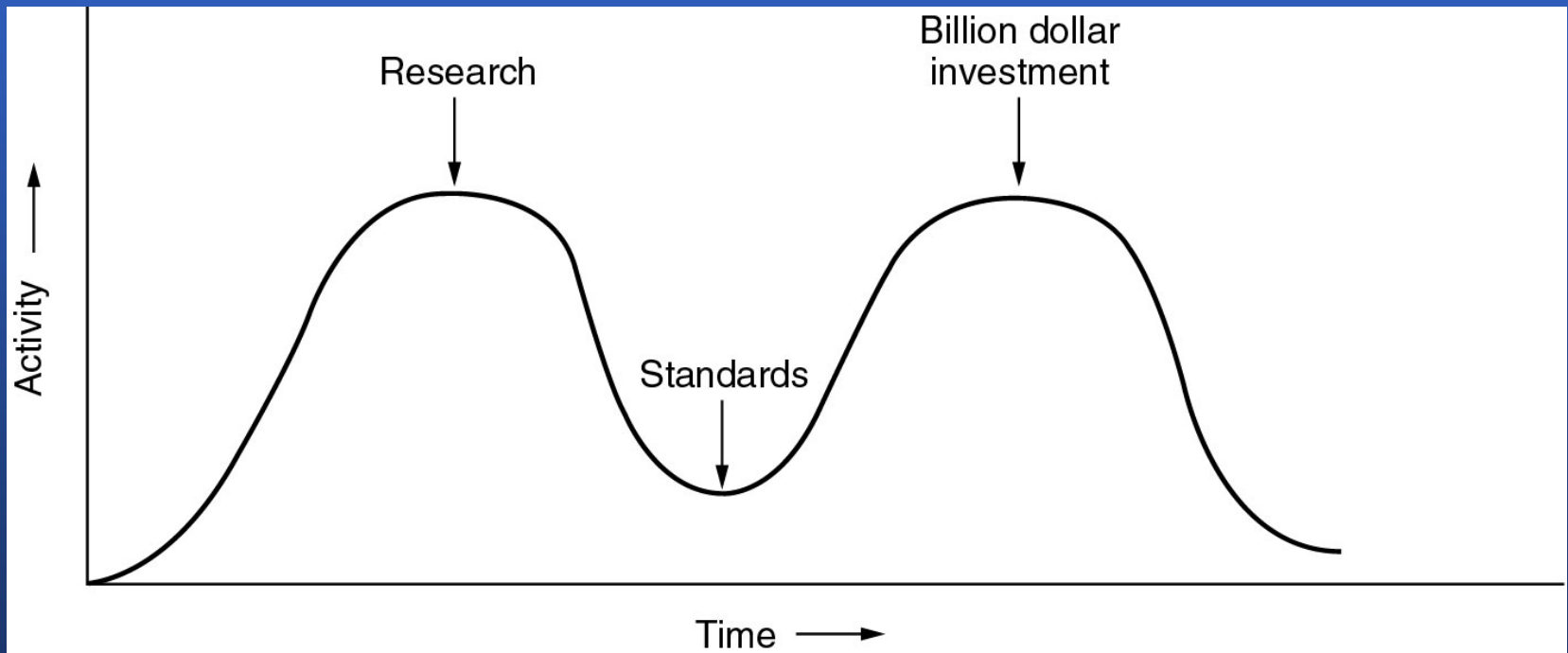
1.4.4 A Critique of the OSI Model and Protocols

- Why OSI did not take over the world
- Bad timing
- Bad technology
- Bad implementations
- Bad politics



Bad Timing

- The apocalypse of the two elephants.



1.4.5 A Critique of the TCP/IP Reference Model

- Problems:
 - Service, interface, and protocol not distinguished
 - Not a general model
 - Host-to-network “layer” not really a layer
 - No mention of physical and data link layers
 - Minor protocols deeply entrenched, hard to replace



Hybrid Model

- The hybrid reference model to be used in this book.

5	Application layer
4	Transport layer
3	Network layer
2	Data link layer
1	Physical layer

1.5 Example Networks

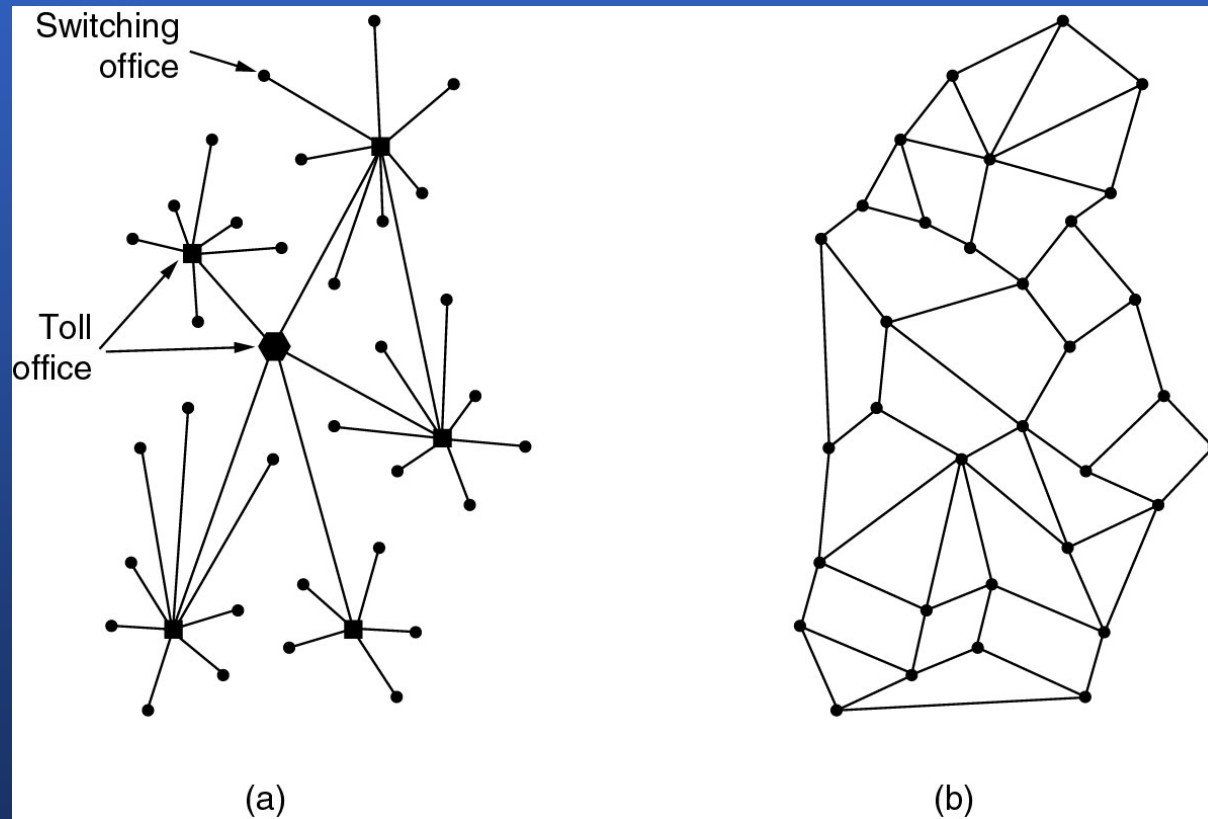
- The Internet
- Connection-Oriented Networks:
X.25, Frame Relay, and ATM
- Ethernet
- Wireless LANs: 802.11



1.5.1 The ARPANET

(a) Structure of the telephone system.

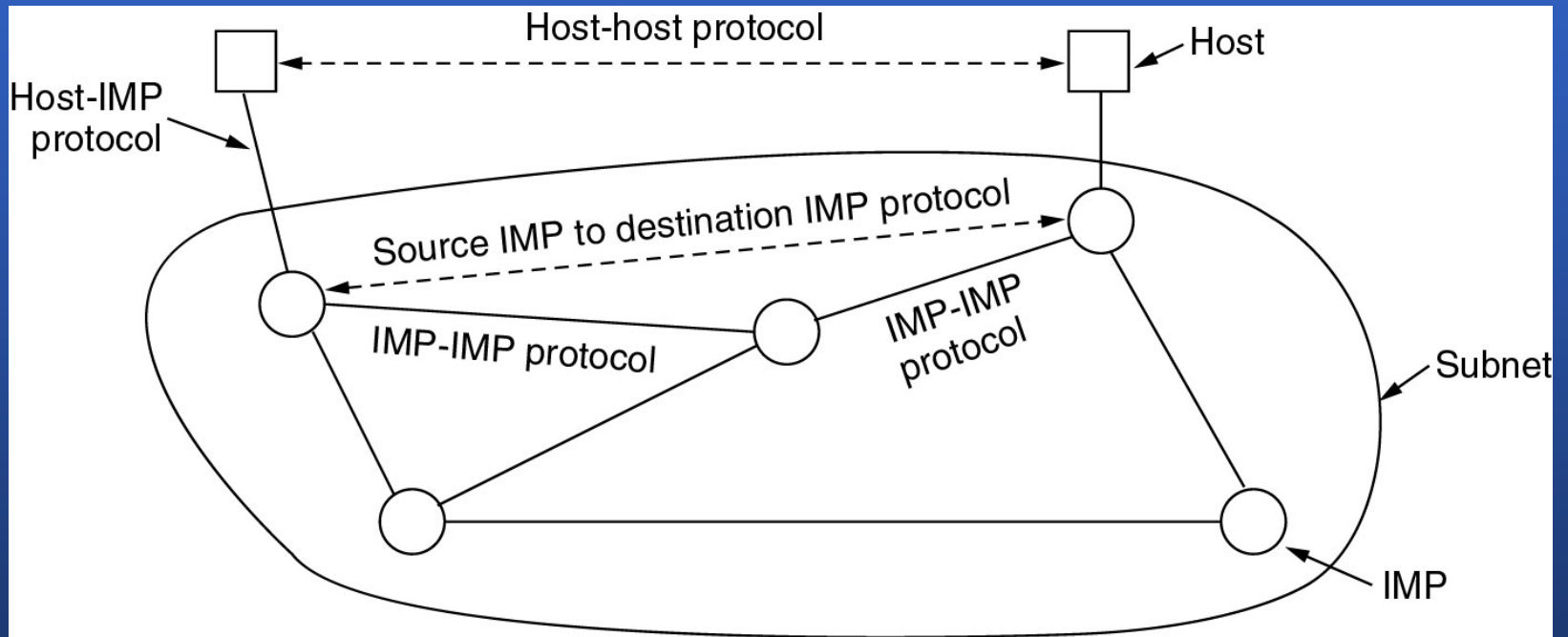
(b) Baran's proposed distributed switching system.



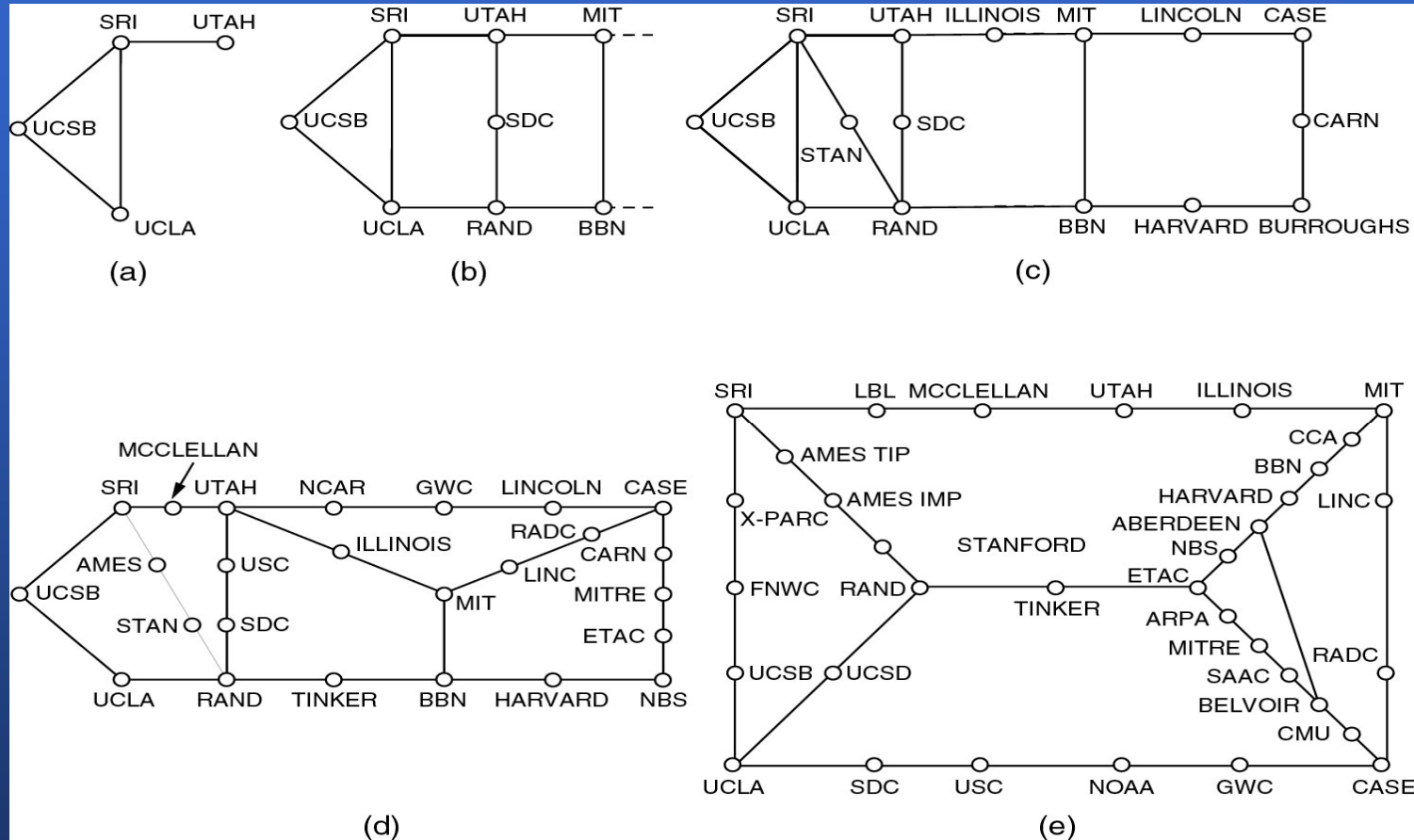
ARPA: Advanced Research Projects Agency of Department of Defense

The ARPANET (2)

- The original ARPANET design.



The ARPANET (3)

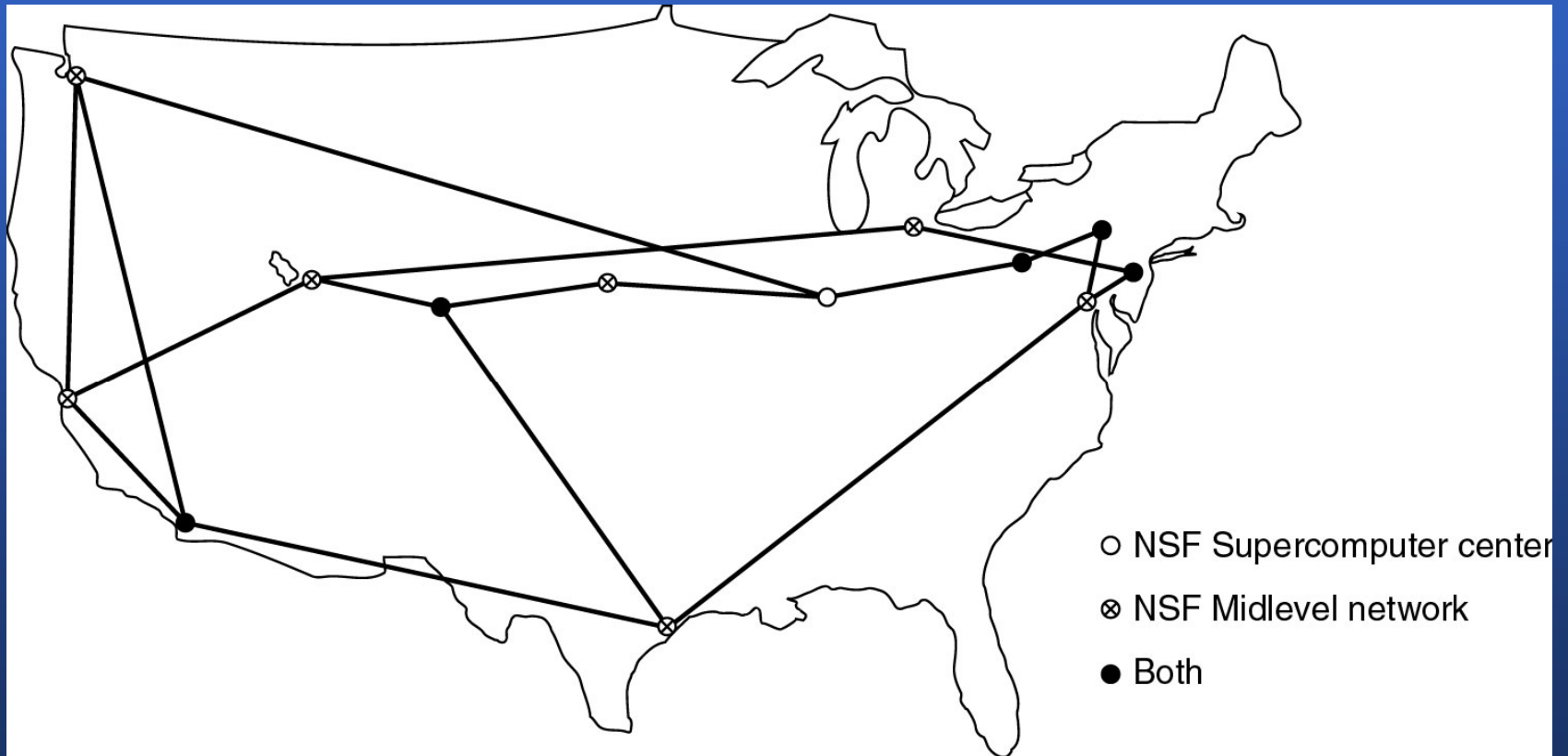


Growth of the ARPANET (a) December 1969. (b) July 1970.

(c) March 1971. (d) April 1972. (e) September 1972.

NSFNET

- The NSFNET backbone in 1988.



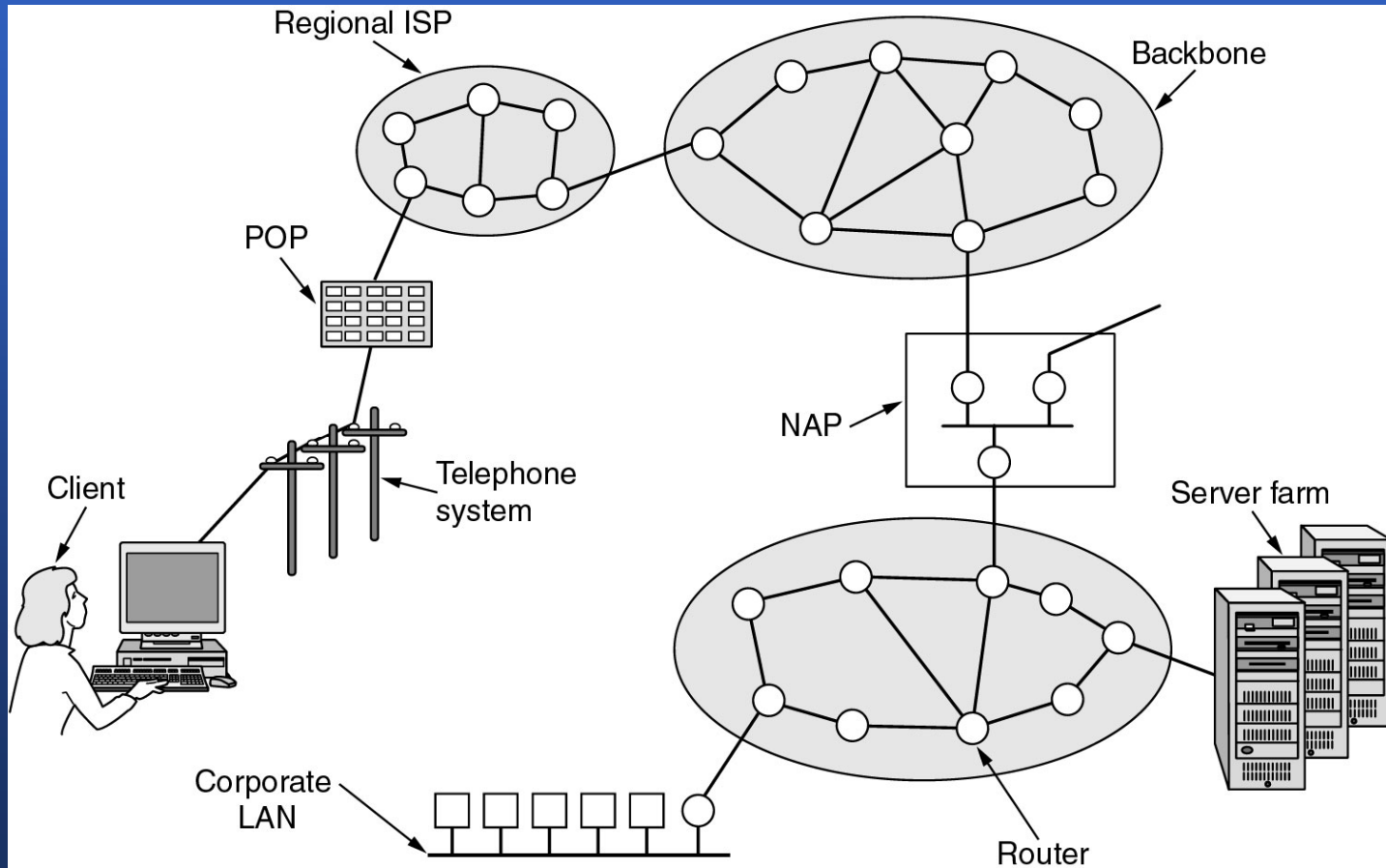
Internet Usage

- Traditional applications (1970 – 1990)
- E-mail
- News
- Remote login
- File transfer



Architecture of the Internet

- Overview of the Internet.



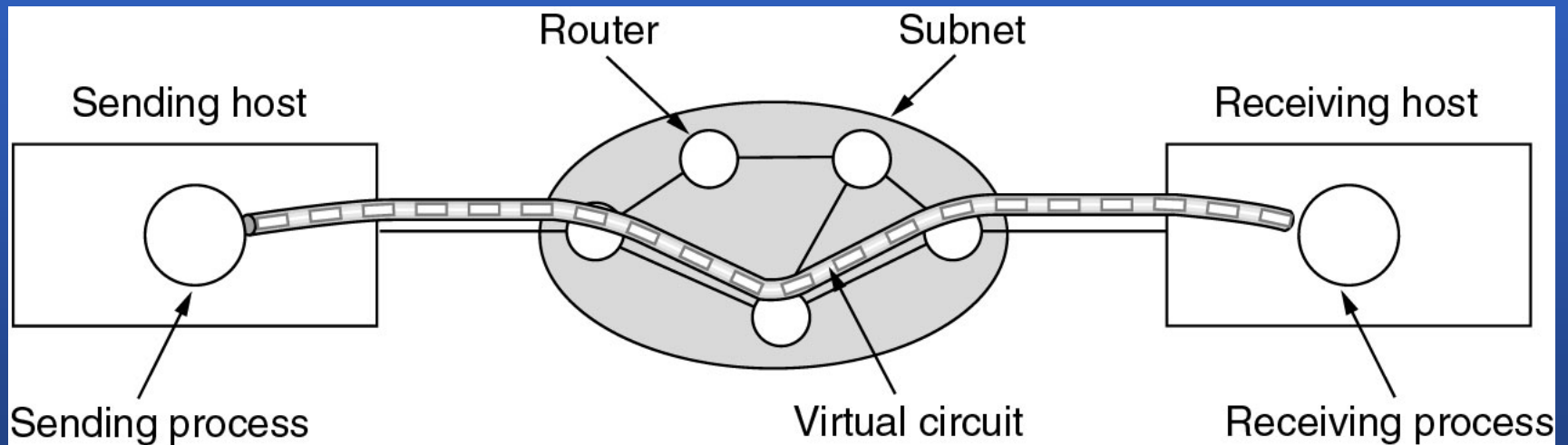
1.5.2 Connection-Oriented Networks: X.25, Frame Relay and ATM

- X.25 -- 1970s
 - First connection-oriented network
- Frame Relay --1980s
 - No error control and no flow control
- ATM (Asynchronous Transfer Mode) – 1990s
 - Cells



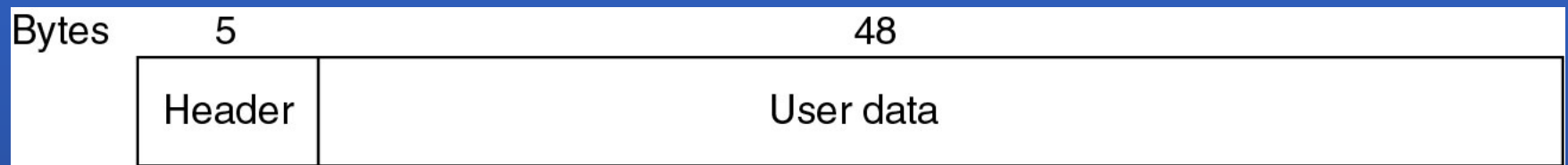
1.5.2 ATM Virtual Circuits

- A virtual circuit.

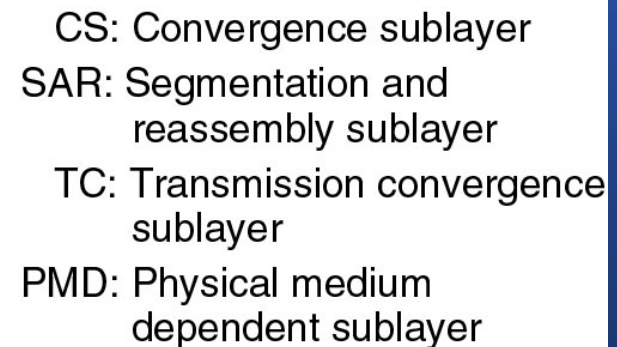


ATM Virtual Circuits (2)

- An ATM cell.



- The ATM reference model.



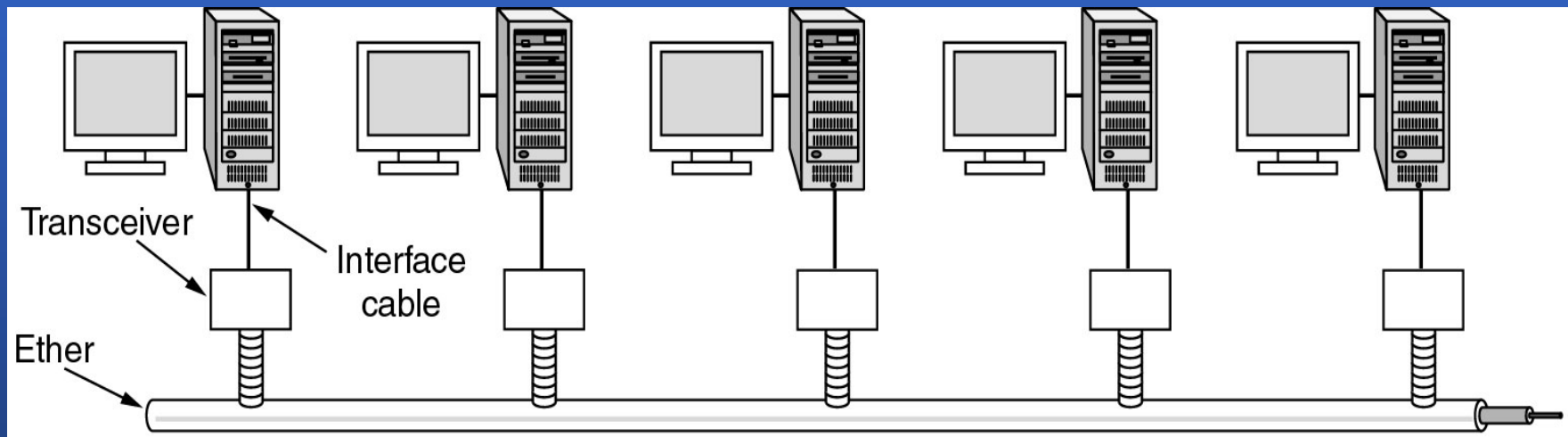
The ATM Reference Model (2)

OSI layer	ATM layer	ATM sublayer	Functionality
3/4	AAL	CS	Providing the standard interface (convergence)
		SAR	Segmentation and reassembly
2/3	ATM		Flow control Cell header generation/extraction Virtual circuit/path management Cell multiplexing/demultiplexing
2	Physical	TC	Cell rate decoupling Header checksum generation and verification Cell generation Packing/unpacking cells from the enclosing envelope Frame generation
1		PMD	Bit timing Physical network access

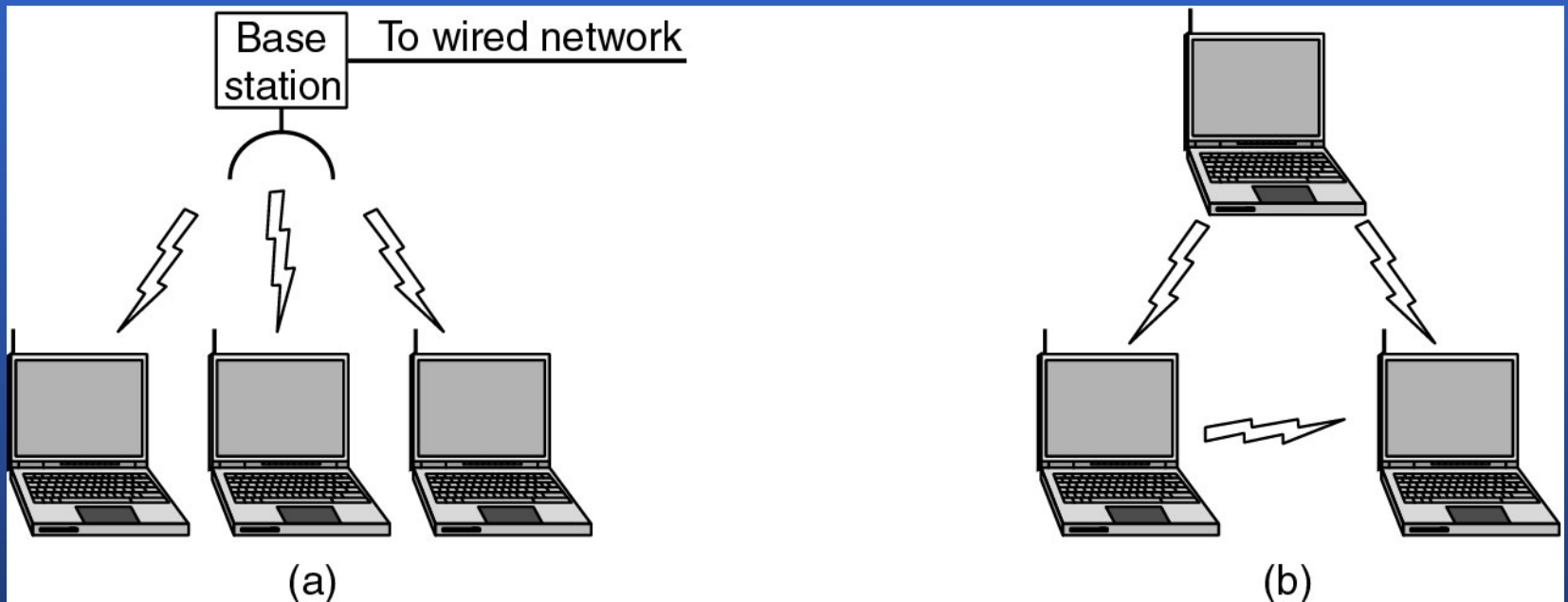
- The ATM layers and sublayers and their functions.

1.5.3 Ethernet

- Architecture of the original Ethernet.



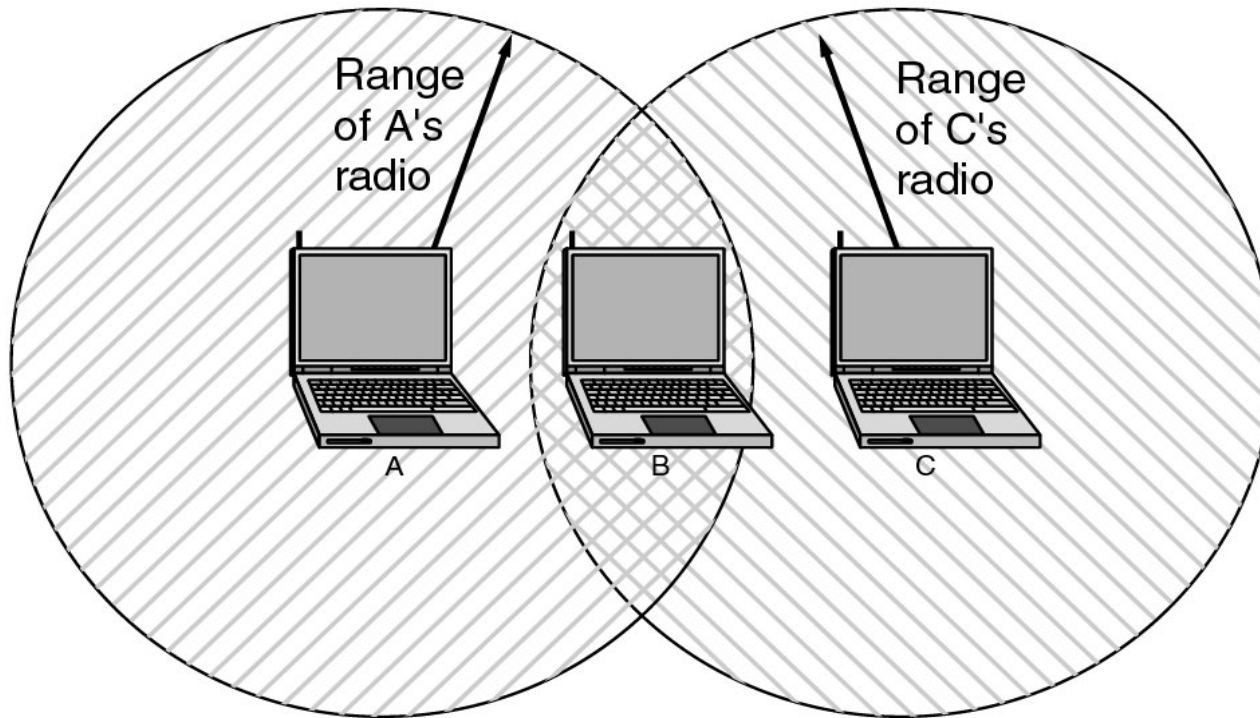
1.5.4 Wireless LANs:802.11



(a) Wireless networking with a base station.

(b) Ad hoc networking.

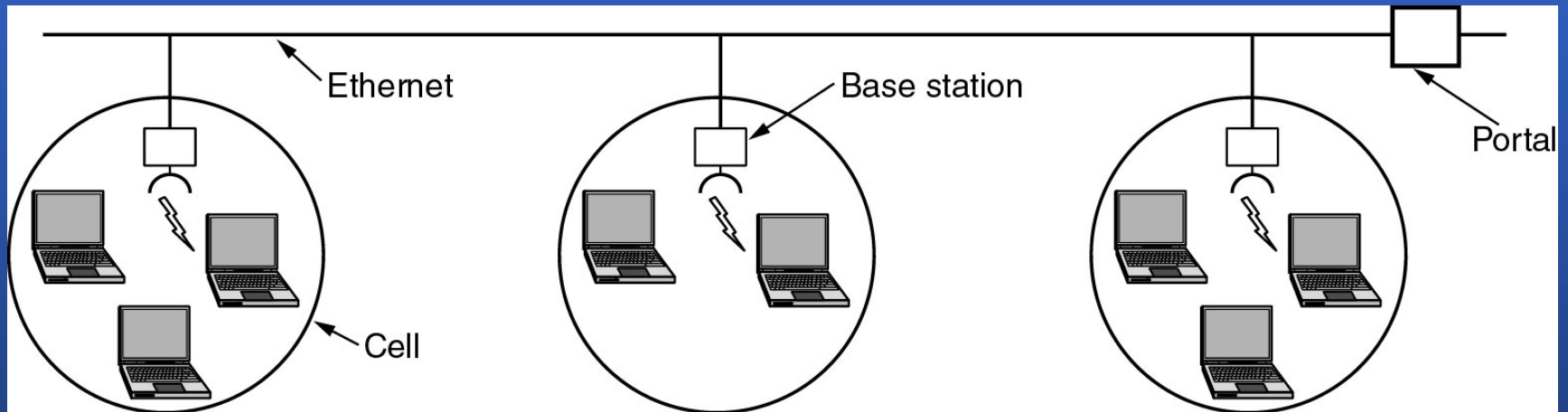
Wireless LANs (2)



- The range of a single radio may not cover the entire system.

Wireless LANs (3)

- A multicell 802.11 network.



1.6 Network Standardization

- Who's Who in the Telecommunications World
- Who's Who in the International Standards World
- Who's Who in the Internet Standards World



1.6.1 Who's Who in the Telecommunications World

- 国际电信联盟 (ITU)
 - 无线通信部门 (ITU-R)
 - 电信标准化部门 (ITU-T)
 - 开发部门 (ITU-D)
- 邮电部 (PTT)
- 公用电信公司, AT&T, Bell



ITU

- Main sectors
 - Radiocommunications (ITU-R)
 - Telecommunications Standardization (ITU-T)
 - (被称为 CCITT, 1953–1993)
 - Development (ITU-D)
- Classes of Members
 - National governments, more than 200
 - Sector members, more than 500, AT&T, Cisco...
 - Associate members, Study Group
 - Regulatory agencies, as FCC



1.6.2 Who's Who in the International Standards World

- 国际标准化组织 (ISO)
 - 标准化程序: 委员会草案 (CD) → 国际标准草案 (DIS) → IS
- 美国国家标准协会 (ANSI)
- 国家标准和技术协会, National Institute of Standards and Technology (NIST)
- 电器和电子工程师协会 (IEEE) 802. x → ISO 8802. x



1.6.2 IEEE 802 Standards

Number	Topic
802.1	Overview and architecture of LANs
802.2 ↓	Logical link control
802.3 *	Ethernet
802.4 ↓	Token bus (was briefly used in manufacturing plants)
802.5	Token ring (IBM's entry into the LAN world)
802.6 ↓	Dual queue dual bus (early metropolitan area network)
802.7 ↓	Technical advisory group on broadband technologies
802.8 †	Technical advisory group on fiber optic technologies
802.9 ↓	Isochronous LANs (for real-time applications)
802.10 ↓	Virtual LANs and security
802.11 *	Wireless LANs
802.12 ↓	Demand priority (Hewlett-Packard's AnyLAN)
802.13	Unlucky number. Nobody wanted it
802.14 ↓	Cable modems (defunct: an industry consortium got there first)
802.15 *	Personal area networks (Bluetooth)
802.16 *	Broadband wireless
802.17	Resilient packet ring

The 802 working groups. The important ones are marked with *. The ones marked with ↓ are hibernating. The one marked with † gave up.



1.6.3 Who's Who in the Internet Standards World

- 因特网活动委员会 (IAB, 1983)→因特网体系结构委员会 (IAB)
 - RFC (Request For Comments, 请求评注)
- 因特网研究特别任务组 (IRTF)
- 因特网工程特别任务组 (IETF)
- 因特网协会 (Internet society)

- RFC→Proposed Standard→Draft Standard→Internet Standard



1.7 Metric Units

- The principal metric prefixes.

Exp.	Explicit	Prefix	Exp.	Explicit	Prefix
10^{-3}	0.001	milli	10^3	1,000	Kilo
10^{-6}	0.000001	micro	10^6	1,000,000	Mega
10^{-9}	0.000000001	nano	10^9	1,000,000,000	Giga
10^{-12}	0.000000000001	pico	10^{12}	1,000,000,000,000	Tera
10^{-15}	0.000000000000001	femto	10^{15}	1,000,000,000,000,000	Peta
10^{-18}	0.000000000000000001	atto	10^{18}	1,000,000,000,000,000,000	Exa
10^{-21}	0.000000000000000000001	zepto	10^{21}	1,000,000,000,000,000,000,000	Zetta
10^{-24}	0.00000000000000000000001	yocto	10^{24}	1,000,000,000,000,000,000,000,000	Yotta

Note:

1. KB,MB,GB for $2^{10}, 2^{20}, 2^{30}$ bytes

2. Kbps,Mbps,Gbps for $10^3, 10^6, 10^9$ bit/sec



1.8 Outline of the Book

CHAPTER 1 INTRODUCTION

CHAPTER 2 THE PHYSICAL LAYER

CHAPTER 3 THE DATA LINK LAYER

CHAPTER 4 THE MEDIUM ACCESS SUBLAYER

CHAPTER 5 THE NETWORK LAYER

CHAPTER 6 THE TRANSPORT LAYER

CHAPTER 7 THE APPLICATION LAYER

CHAPTER 8 NETWORK SECURITY



Exercises

- 11, 13, 18, 22, 27

